Rahul Yedida

hello@ryedida.me

Website :: GitHub :: LinkedIn :: Google Scholar :: Google Developers

+1 (206) 660-7542

Employment

LexisNexis Legal & Professional — Raleigh, NC

Senior Data Scientist II — *Oct 2026 - Present*

- Fine-tuned small structured-decision models to match LLM-baseline performance (macro-F1 92.7%) at 177 ms median latency on-device, after benchmarking 15 models across 6 tasks.

Senior Data Scientist I — *May 2024 - Oct 2026*

- Technical lead for an agentic legal drafting system; led its migration from a RAG pipeline and own its ongoing development.
- On a 3-person team, increased the share of drafts rated suitable for users by attorney reviewers from 40% to 86% for complaints and 59.2% to 76% for motions.
- Reduced customer-facing product runtime by 24.8% through LLM call profiling and parallelization.
- Built a contrastive-learning filter using InfoNCE with alignment and uniformity losses to remove unrelated legal passages from search results; median filtering latency of 12 ms across 40 queries.
- Built team developer tooling: custom Python lint rules beyond `ruff`'s set, enforced in CI; a Rust MITM proxy with request replay that teammates use to debug without waiting on upstream responses; and a web UI for visualizing intermediate agent steps.
- Mentored an intern on a large-scale, full factorial experiment investigating Pareto-optimal configurations for draft quality and latency.
- Profiled high pod memory usage and introduced lazy module loading, reducing time at peak memory from 86.4% to 68% of runtime.
- **Technology:** Python, Litestar, Google ADK, React, Rust, TypeScript, Tailwind

Amazon — New York, NY / Bellevue, WA

Software Dev Engineer Intern — *May 2023 - Aug 2023*

- Implemented and shipped profile locks for Prime Video on Echo Show devices.
- **Technology:** React Native, TypeScript

Software Dev Engineer Intern — *May 2022 - Jul 2022*

- Developed a full-stack system to publish announcements in scorecards used by delivery service partners.
- **Technology:** React/Redux, TypeScript, Redux Saga, DynamoDB, Java Spring

Relevant Projects

pi-permission-system (maintained fork) — May 2026 - Present

TypeScript, Node.js — *GitHub*

Maintain a permission-system fork for the Pi coding agent. Redesigned shell permissions using grammar-based analysis of pipelines, substitutions, and redirections, with protected-path checks and argument-aware approval rules. Extended enforcement to background commands and fixed concurrent subagent approval handling.

Q&A System for Zotero — Oct 2024 - Present

Rust, SQLite, LanceDB — *GitHub :: crates.io*

Built an academic question-answering and semantic search system over a Zotero library. Developed a Rust RAG library with configurable LLM, embedding, and reranking providers, tool calling, and limits on concurrent embedding requests. Wrote a custom PDF parser for academic papers (median throughput ≈ 2,400 pages/s across 11 PDFs; M4 Pro).

JournalBear — Jan 2017 - Jul 2026

TypeScript, React, Zustand, Electron, SwiftUI — *GitHub :: App Store :: Softpedia*

Built and shipped a journaling app with AES-256 encryption, an Electron/React desktop client, and native SwiftUI apps for iOS and macOS.

The 1 Billion Row Challenge (1BRC) — Feb 2026 - Apr 2026

Zig — *GitHub*

Wrote a Zig implementation for the 1BRC, running in 1.20s (M4 Pro, 14 core) using mmap, open addressing (Wyhash + Robin Hood probing), and packed SIMD.

Programmable Resumes — Aug 2023 - Mar 2025

Python — *GitHub*

Developed a resume specification language and Python compiler that generates PDFs from structured resume data via LaTeX. Supports tag-based role filtering, two LaTeX templates, and parallel builds. Wrote this resume using this tool.

Skills

Languages: Rust, Python, TypeScript, Java, C++, Gleam

ML/AI: PyTorch, Keras, Hyper-parameter optimization, RAG, LLM evaluation, Agentic systems, Google ADK

Frameworks: React, FastAPI, Litestar, FastAPI

Databases: LanceDB, SQL, MongoDB, DynamoDB

Cloud: Google Cloud (Compute Engine, Cloud Storage, Vertex AI), AWS (EC2, S3, DynamoDB)

Education

North Carolina State University — Raleigh, NC
Ph.D. Computer Science — *Aug 2019 - Jul 2024*

- Advisor: *Dr. Tim Menzies*
- Dissertation: *Guidelines for the Application of Neural Technologies in Software Analytics (or: How to Do More with Less in SE)*

PES University — Bangalore, India
B.E. Computer Science – Advisor: Dr. Snehanshu Saha — *Aug 2015 - May 2019*

Honors

Distinguished Reviewer, ICSE 2026 — Mar 2026
Google Developer Expert - ML and GCP — Mar 2025 - Present
Google Cloud Champion Innovator - Cloud AI/ML — Oct 2022 - Mar 2025
Google Cloud Research Innovator — Feb 2022